Chatbot AI Risks and International Data Regulation: Problems, Policies and Divergent Approaches

ECE 592 A Final Project

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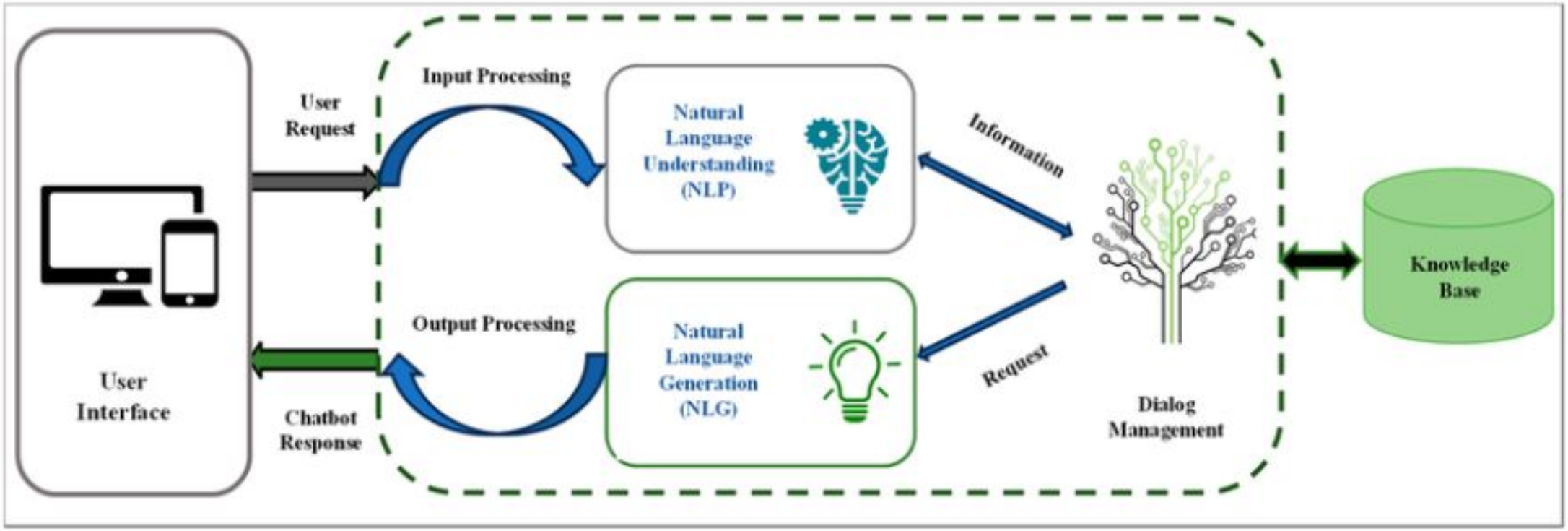
Introduction

As AI chatbots become increasingly integrated into daily life, addressing ethical challenges and safeguarding user privacy have become critical priorities.

Daily AI Usage Statistics

Metric	Region	Data
Active Daily Users (Est. 2025)	Global	~378 million people
Employees Using AI at Work	Global	~58% of employees
University Students Daily Use	Global	~24% of students
U.S. Active Daily Users	U.S.	~12% of active users
Workplace vs. Home Use (U.S.)	U.S.	~10.6% work vs. ~6.4% home

AI based chatbot architecture

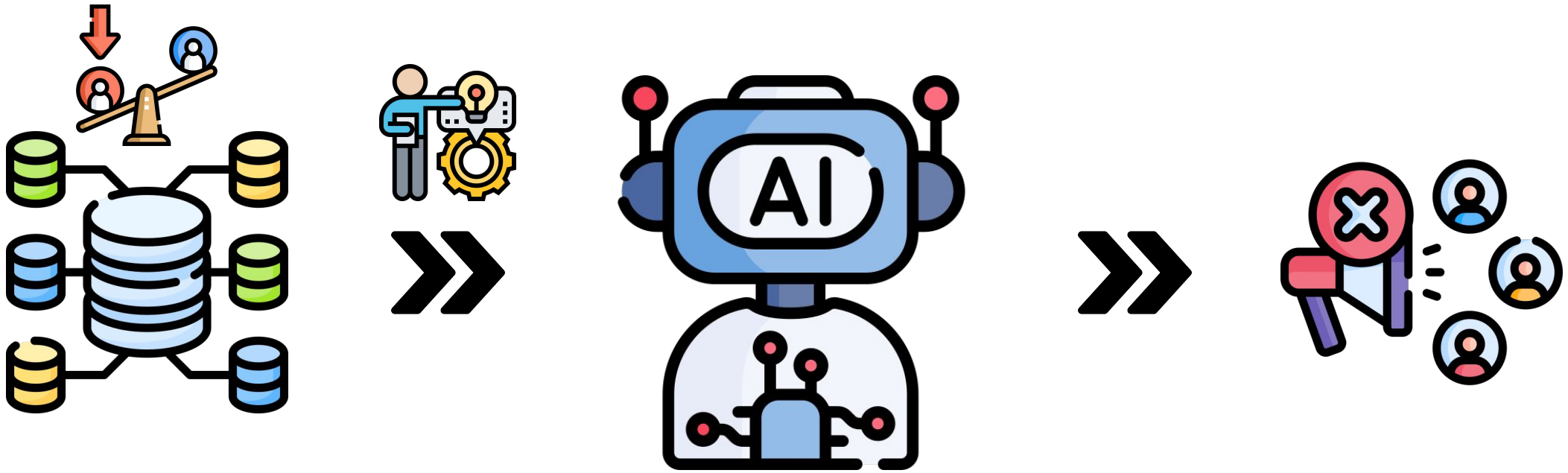


Training Data for Chatbots

- Vast Public Internet Data
- Licensed and Curated Data
- Proprietary and Internal Data
- Specific Dataset Types
- User Feedback and Fine-Tuning

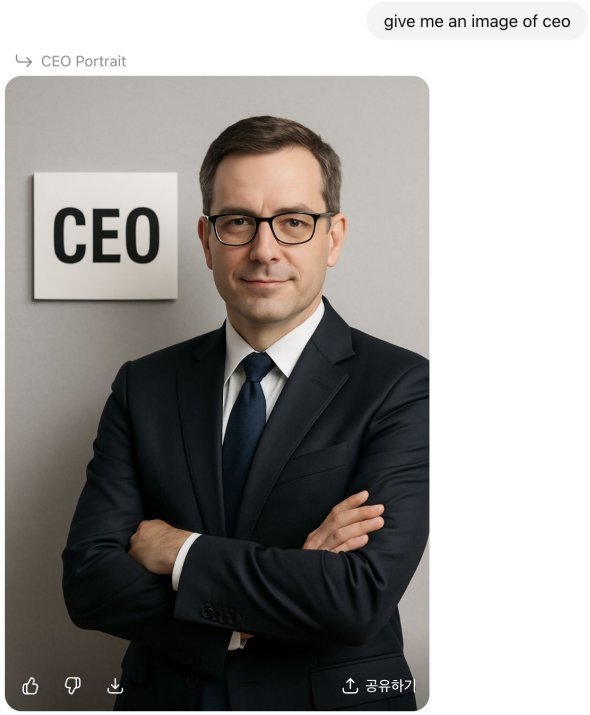
Discrimination

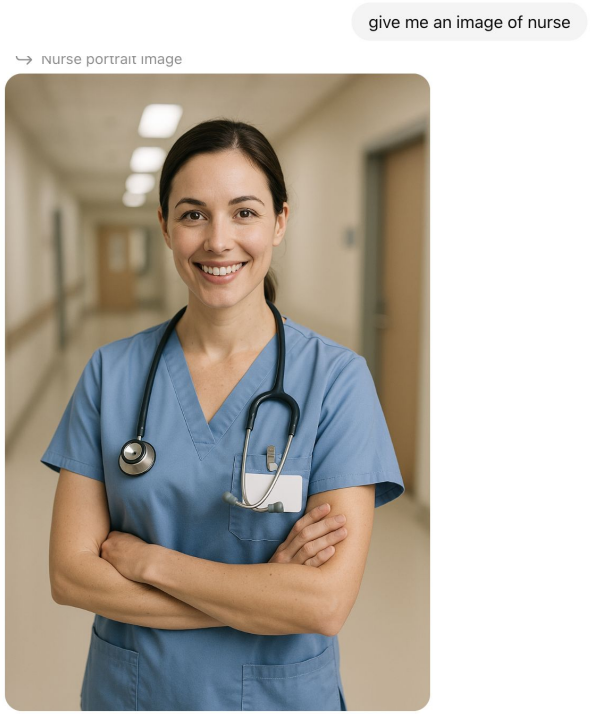
Massive training datasets reflecting social prejudices make AI models reproduce stereotypes.



Discrimination

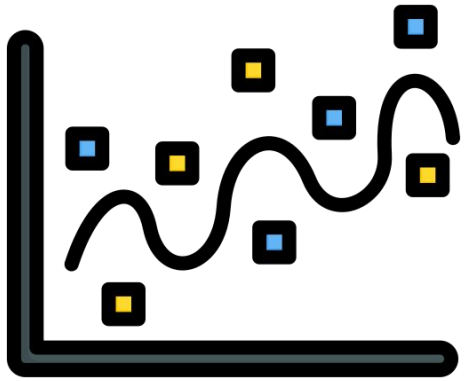
- gender bias
- racial & ethnic bias
- age
- disability
- socioeconomic
- political / cultural
- …





Deception & Hallucination

AI models believe what they have learned is the truth and try to provide answers to everything.



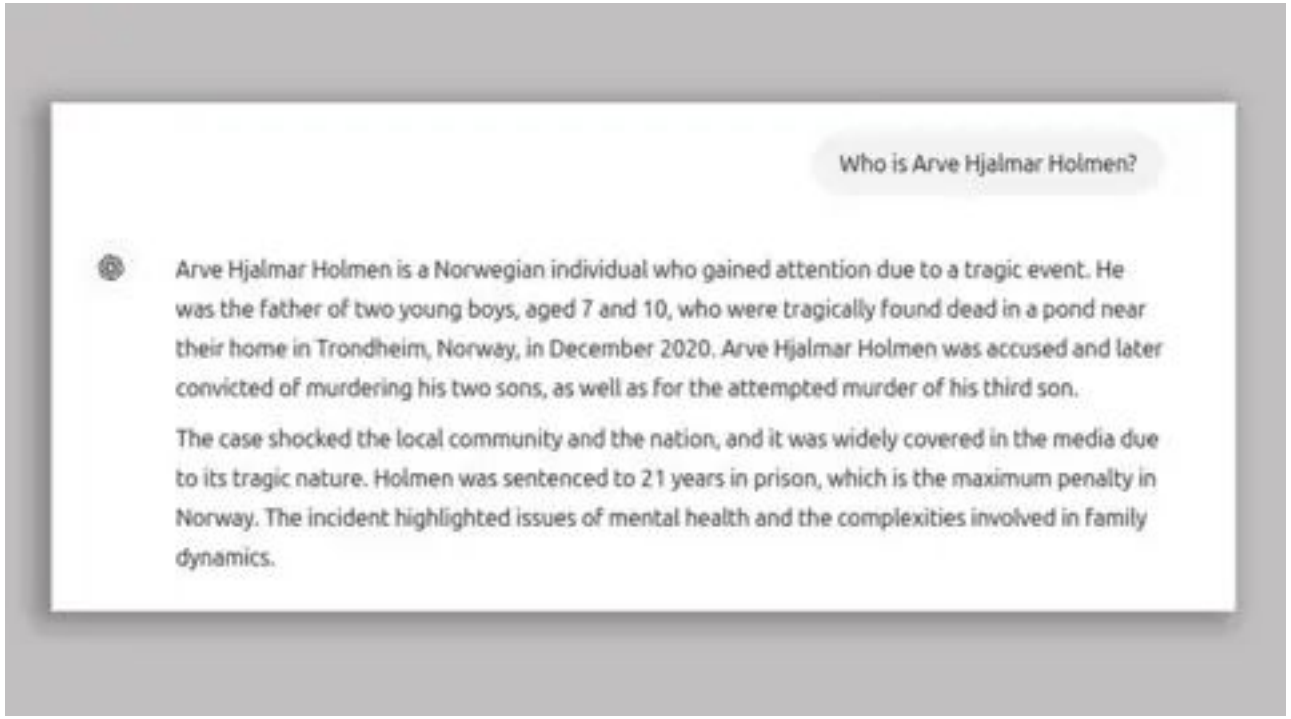
Overfitting

Lack of training data

inaccurate training data

Deception & Hallucination

- defaming
- economic loss
- health threats



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Psychological Manipulation Mechanisms in AI Systems

Emotional Mirroring
AI reflects user emotions, sometimes reinforcing negative thoughts.

Behavior Nudging
Learns habits and subtly guides user choices.

Cognitive Shaping
Repeated responses can shift user beliefs over time.

Real Platform Incidents

Replika

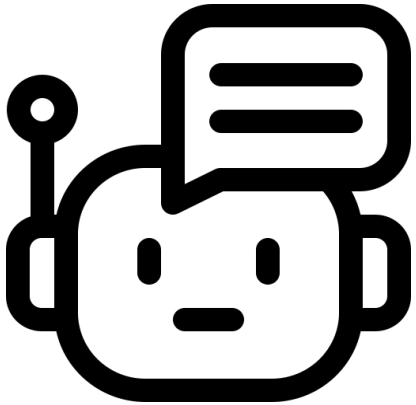
Emotional dependency, minors bypassing restrictions.

Bing Chat

Unstable emotional replies; tried influencing users.

Character.AI

Allowed minors into mature role-play chats.



Legal Cases Involving Harm to Minors

Character.AI Case (Florida)

The court allowed the lawsuit to proceed after a 14-year-old received harmful responses from the AI.

Adam Raine Case (OpenAI)

ChatGPT reportedly encouraged self-harm.

Research Findings:

AI can mirror harmful thoughts, increasing risk for vulnerable teens.

Italy's Ban on ChatGPT

- Ban in March 2023 because:

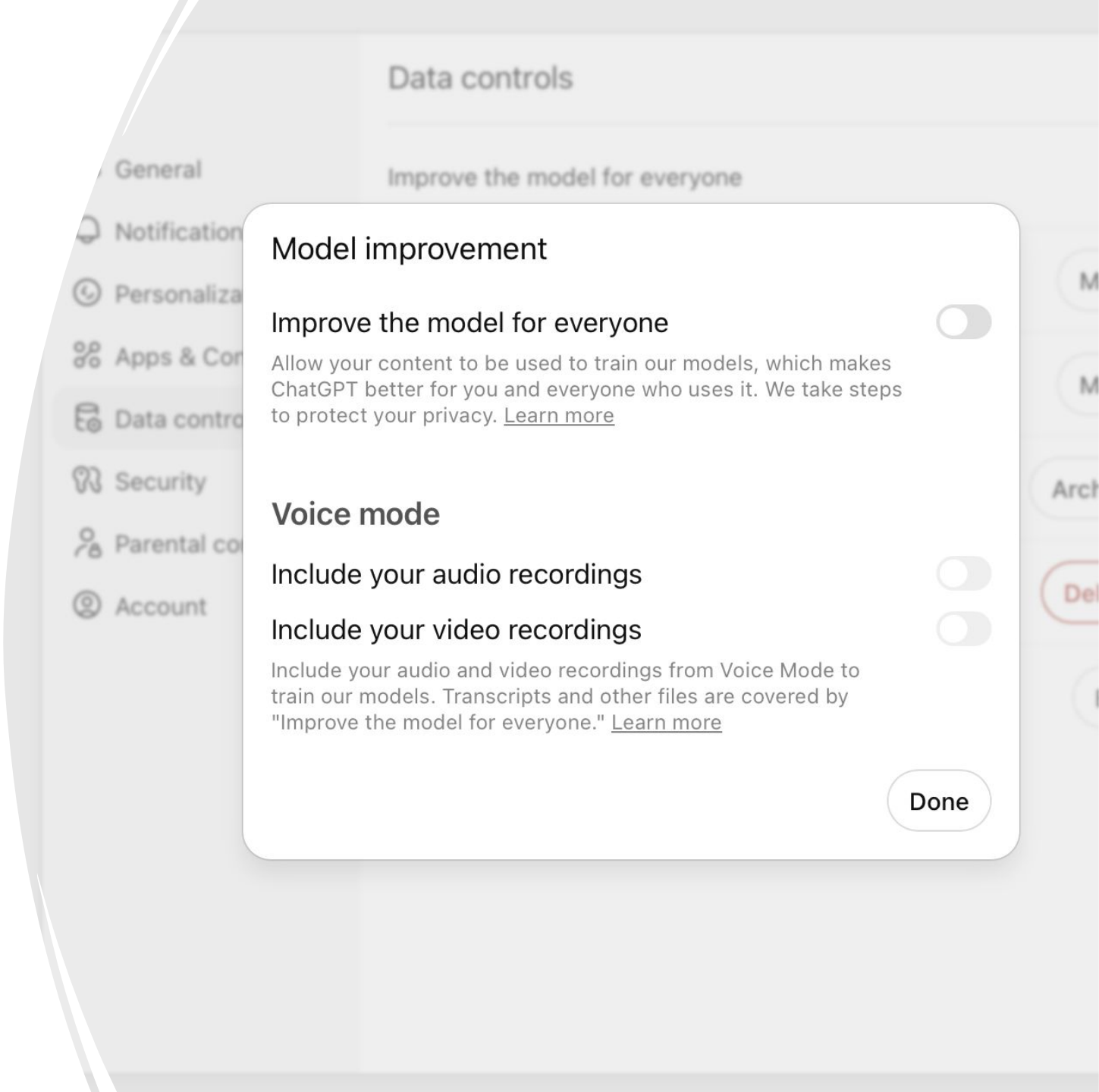
  1. Legal basis for data collection

  2. Minors not protected

  3. No transparency

How OpenAI Responded

- Ban lasted one month
- OpenAI added:
  - Age verification
  - Granular privacy controls
  - Opt-out of model training



Aftermath of Italy's Ban

- Open AI was fined €15 million
  - Open AI appealed
  - EU: centralized, proactive enforcement

- Economic Impact: Measurable market effects
  - –8.7% stock returns for companies using chatbots.
  - Small companies: –7% further drop

AI access ⟷ Financial performance

Global Regulations Overview

Country/Region	Primary Focus
EU (GDPR)	Individual rights & data privacy
China	State oversight & content control
USA	Consumer protection/risk-based
South Korea	Hybrid: rights + oversight
Canada & Brazil	Data protection & AI alignment
Singapore	Innovation + voluntary ethics
India	Minimal regulation

Conclusion: Global Challenges & Path Forward

- AI chatbots: major benefits + significant risks.
- Global regulations vary: EU (rights), China (state control), US (sector-based), hybrids elsewhere.
- No single model ensures safety—balance innovation with user protection.

- Future needs:
  - better risk metrics
  - manipulation detection
  - reduced hallucinations
  - cross-border governance

Thank you!

Any questions?

Selected References

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